
Fixing the Background: Segmentation-Guided Feature Refinement for Multiple Object Tracking

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Abstract: Tracking-by-detection algorithms in multiple object tracking systems rely on bounding boxes for object localization and data association. Bounding boxes also contain extraneous background information, which contaminates appearance features, thereby increasing false or missed detections. This paper proposes a multiple object tracking framework that integrates segmentation-based background removal into the tracking pipeline. Mask R-CNN is used to eliminate background clutter within the bounding boxes, producing refined detections. Deep SORT uses a Kalman filter for motion modeling and, in parallel, a convolutional neural network extracts appearance features from these refined detections. Data association is performed by evaluating the similarity between predicted tracks and detections using both motion and appearance features. The final matching between tracks and detections is performed using the Hungarian algorithm, enabling improved object detections over time. The robustness of data association is enhanced in challenging situations, such as occlusion and clutter, by improving appearance features. The approach demonstrated improvement in the overall tracking performance on the MOT16 dataset, resulting in reduction in false or missed detections, and an increase in multiple object tracking accuracy (MOTA).

Keywords: Multiple Object Tracking, Segmentation-based Tracking, Deep SORT, Background removal, Mask R-CNN

1. Introduction

Multiple Object Tracking (MOT) [1], [2] is an eminent paradigm in computer vision with a wide range of applications in surveillance systems, autonomous driving, robotics [3], traffic monitoring [4], and behavior analysis [5]. It focuses on calculating and preserving consistent identities of multiple objects across consecutive video frames. Data association [6] in real-world scenarios, such as scale variations, occlusions, and crowded scenes, becomes suboptimal, thereby reducing the tracking accuracy and increasing false and missed detections. In MOT [1], tracking methods can be categorized into detection-free tracking (DFT), detection-based tracking (DBT), and detection-tracking hybrid methods [2]. Detection-free methods transmit states of manually initialized objects using motion or appearance models. They are not applicable for long-term applications and struggle with initialization errors resulting from manual initialization. Hybrid models enhance tracking accuracy by creating an integrated framework that combines detection and tracking, but often increase computational cost. Detection-based tracking methods detect objects first and then associate these detections consistently across frames. The decoupling of tracking and detection enables real-time performance and optimized tracking.

Detection-based tracking has become predominant in MOT [1], primarily due to the strong performance of deep learning-based object detectors [2]. Object detector forms the backbone of tracking-by-detection systems. In early MOT pipelines, proposal-based detectors such as Fast R-CNN [7] and Faster R-CNN [8] were used for object detection [9]. Although proposal-based detectors provide accurate detections, they employ two-step detections and are thus computationally expensive. Subsequent grid-based detectors, such as single-step YOLO [10] models from YOLOv3 [11] to YOLOv8, and YOLOX [12] redefined detection by introducing anchor-free detection and decoupled heads [13]. YOLO variants are preferred in MOT systems due to improved efficiency and real-time performance. Unlike grid-based detectors, the latest transformer-based detectors, such as DETR [14], eliminated both hand-crafted anchors and non maximum suppression (NMS). They enabled end-to-end optimization by using bipartite matching and object queries. Their relevance in MOT is further strengthened by DINO [15] and RT-DETR [16], which improve convergence speed along with real-time capability.

Various robust and efficient tracking-by-detection MOT approaches have been developed over the years. One of the earliest MOT methods, Simple Online and Realtime Tracking (SORT) [17], integrated a Kalman filter for motion prediction with the Hungarian algorithm employing IoU-based similarity for data association. SORT is a lightweight online tracking framework that relies mainly on the Jaccard index (IOU) and motion features for association, making it susceptible to appearance confusion and occlusions. Affinity-based methods [6] learn similarity functions between detections to improve tracking instead of solely relying on metrics such as IoU or Mahalanobis distance. For instance, Deep SORT [18] significantly enhanced accuracy during occlusions by incorporating appearance descriptors learned through a convolutional neural network (CNN) [19]. Nevertheless, Deep SORT still depend on feature extraction from bounding boxes [20], which often include background information that can impact association accuracy and degrade the quality of appearance features in crowded scenes. Furthermore, its tracking performance is strongly impacted by re-ID quality [21] and Kalman filter assumptions.

Heuristic-based frameworks, such as ByteTrack [22] and BoT-SORT [23], have improved association strategies. ByteTrack [22] enhances association by utilizing high-confidence and low-confidence scores. BoT-SORT [23] incorporates camera motion compensation (CMC) for motion modeling along with refined appearance based association strategies. StrongSORT [20] also enhances DeepSORT by integrating camera motion compensation, improved appearance modeling, and global association. Beyond these heuristics-based methods, motion modeling has also evolved beyond classical Kalman Filter assumptions. OC-SORT [24] improves motion modeling through observation-centric online smoothing (OOS), where, through interpolation, missing trajectories are reconstructed, which improves performance in cases of partial occlusions and fast motion. Deep OC-SORT [25] further improves motion modeling by updating appearance features based on detection scores, making trackers more robust to noisy detections. Traditional motion models are replaced by more advanced transformer-based methods such as MotionTrack [26], ETTrack [27], and TransTrack [28] that specifically handle occlusions and abrupt motion. Transformer-based models achieve improved performance in challenging environments, such as sports [29]. TWIX [30] further improves tracking by removing the need for handcrafted motion assumption and learning association from spatio-temporal relationships using transformer-based learning. QDTrack [31] and SMILETrack [32], appearance-based models use transformer-based embeddings and dense contrastive learning to improve tracking in similar object scenes.

Offline tracking methods [33] in MOT formulate as a global graph optimization problem, where associations are edges and detections are nodes. MPNTrack [34] and Sushi [35] are graph-neural network-based methods [36] which refine appearance and motion by message passing over these graphs. These methods are computationally expensive [1] and unsuitable for real-time applications, but can achieve high accuracy. Methods such as CoNo-Link [37] reduce complexity and improve scalability while maintaining accuracy by learning to prune the association graph.

End-to-end detection-tracking hybrid transformer-based methods, such as TrackFormer [38], MOTR [39], [40], and DETR [14], redefine MOT as a sequence modelling. These methods jointly perform detection and association by generating track queries that carry object identity across frames and use attention mechanisms [41]. MOTR updates tracks queries iteratively. Despite this unified approach, these methods still rely on bounding-box representations [1] which degrade appearance features by including background information inside bounding boxes. These methods [38], [39], [40], [14] underperform tracking-by-detection methods in real-time as they are computationally expensive.

Despite these advances in MOT, reliance on bounding-box-based feature extraction is a common limi-

tation in efficient tracking-by-detection methods. In this work, we address this limitation by proposing a segmentation-based tracking framework to improve appearance features. This framework integrates Mask R-CNN-based instance segmentation [42] into the tracking pipeline of the Deep SORT. The segmentation masks are used to subtract the background regions within the bounding boxes. Deep SORT uses these refined detections for data association. Thus, improving feature extraction and reducing missed and false detections in occluded and cluttered environments. Deep SORT is chosen as a standard algorithm to prove the concept of our idea. This technique has potential to be extended to its recent variants and other advanced detection-based multiple object tracking algorithms.

2. Proposed Method

Deep SORT [18] and Mask R-CNN [42] are integrated to form a segmentation-based pipeline to improve tracking performance in crowded environments. Deep SORT performs multiple object tracking through motion and appearance-based data association, whereas for object detection and instance segmentation, Mask R-CNN is used. With the objective of improving appearance during tracking, this architecture is designed to suppress the background information within the bounding boxes. Pre-trained on the COCO2017 dataset [43], Mask R-CNN is employed for human segmentation. Segmentation masks generated by Mask R-CNN during inference are leveraged to subtract the background information inside the detected bounding boxes. Subsequently, Deep SORT uses these refined detections as input for object tracking.

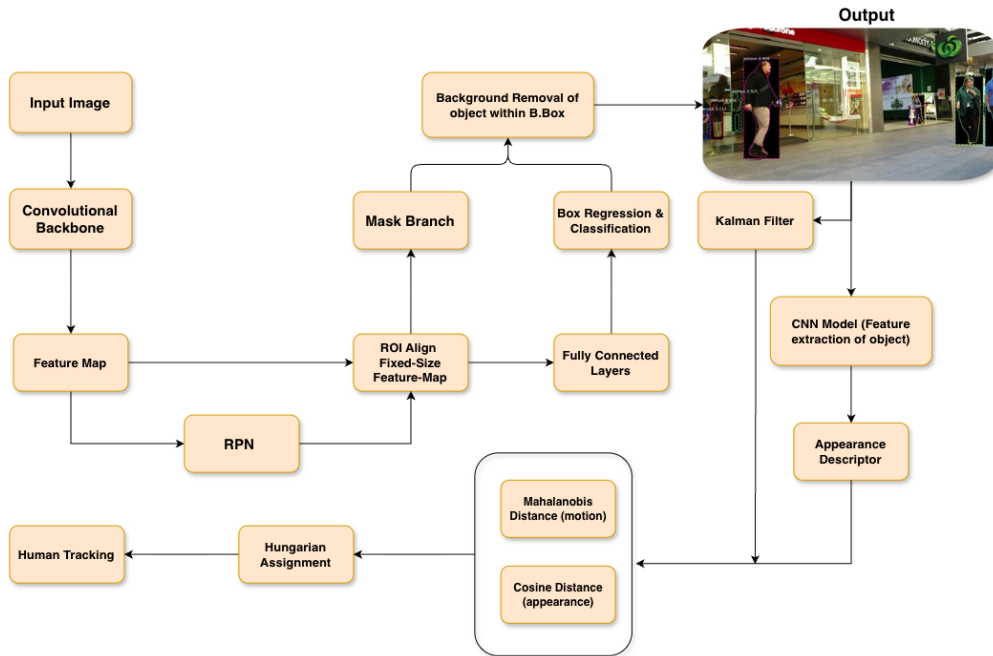


Figure 1. Proposed Technique

The integration of Mask R-CNN into Deep SORT forms a robust tracking framework. Our system aims to improve the tracking performance of Deep SORT by passing segmented human from Mask R-CNN to the appearance descriptor. See Fig. 1 for the overall workflow of the proposed method. The architectural details of the proposed technique are provided in the subsequent sections.

2.1. Detection and Segmentation

Mask R-CNN [42] is the extension of Faster R-CNN [8]. Faster R-CNN only predicts bounding boxes and object class labels, whereas Mask R-CNN extends it by adding a branch of pixel-level segmentation. Mask R-CNN generates segmentation masks for detected objects. Its architecture includes ResNet [19] and Feature

Pyramid Network (FPN) [44] as the backbone for feature extraction. Furthermore, the extracted feature maps are used by the Region Proposal Network (RPN) [8] to generate candidate object regions. Extracted features from the region of interest are processed through RoI pooling for mask prediction, bounding box parameters, and object classification. A fully convolutional layer (FCN) [45] is employed for segmentation mask generation. The mask generation stage involves creating a binary mask that shows the pixels belonging to the objects in the image. After obtaining the mask, the bounding box parameters are obtained from the Region Proposal Network (RPN).

The center point of the bounding box represents the midpoint of the rectangle enclosing the object. The height and width of the bounding box define the dimensions of the rectangle enclosing the object. We have calculated the coordinates of the rectangle surrounding the object by using the center point, height, and width of the bounding box. Afterwards, we take the previously generated mask and apply it within the estimated rectangle. This involves retaining only the pixels within the rectangle that correspond to the object, effectively removing the background inside the bounding box. Fig. 2 shows a sample output of the background removal.



Figure 2. Background Removal

Mask R-CNN is robust in performing object detection and instance segmentation tasks, which is selected as the detector for Deep SORT. Mask R-CNN performs both instance segmentation and object localization simultaneously, unlike other conventional object detectors, making it suitable for tracking applications.

2.2. Tracking

Deep SORT [18] is the extension of SORT [17] and is employed as the tracking module in the proposed framework. It extends SORT by incorporating appearance modeling, reducing missed and false detections by a big margin. For tracking and handling, an eight-dimensional state space is used in Deep SORT. For each track k , the number of frames since the last successful measurement association is denoted by ak . During Kalman filter predictions, ak is incremented and then reset to zero. Tracks that reach a predefined maximum age, A_{max} , are removed from the track set since they are assumed to have disappeared from the scene. The additional track hypotheses are generated with each unmatched detection. These new tracks are believed to be in their preliminary stages for the first three frames. At each timestep within this period, a successful measurement association is expected. Tracks are removed if they cannot be linked successfully to a measurement within the first three frames.

Data association between Kalman states and newly collected detections is solved by building an assignment problem through the Hungarian algorithm. By utilizing a hybrid of two complementary metrics, motion and appearance data are employed to address the issue. The Mahalanobis distance is used between

anticipated Kalman states and recently acquired measurements to include motion information. The uncertainty in state estimation is handled by the Mahalanobis distance, which estimates how far the average track location is from the detection. Rapid displacement can be produced by a real-time camera motion, which makes the Mahalanobis metric unreliable for tracking objects during occlusion. To solve the entity association problem, the appearance descriptor is computed through a pre-trained CNN [19] for each bounding box detection d_j , which in our case has the background removed. For short-term projection, object placement based on motion is given by the Mahalanobis distance metric. However, cosine distance considers physical appearance, which enables the technique to determine the identities after occlusions. Successful use of the technique requires offline training of a discriminative feature embedded using basic nearest neighbor queries and without further metric learning. To achieve this, a CNN is trained on a large-scale dataset of pedestrian images [46] for person re-identification, making it suitable for deep metric learning for human tracking. The dataset contains 1,100,000 training images from 1261 humans.

3. Results and Discussion

The performance of the proposed architecture is evaluated on the MOT16 challenge [47] and compared against the standard approach without background removal. In Tables 1 and 2, the quantitative results are stated using benchmark tracking metrics, including multiple objects tracking accuracy (MOTA), multiple objects tracking precision (MOTP), fragments, mostly tracked targets (MT), and identity switches (IDSW).

Table 1. Tracking Performance Comparison between Original and the Proposed Segmentation-Guided Deep SORT

Sequence	MOTA $\uparrow$		MOTP $\uparrow$		IDSW $\downarrow$	
	With BG	Without BG	With BG	Without BG	With BG	Without BG
MOT16-02	29.0	28.8	75.2	75.1	107	111
MOT16-04	46.6	46.7	78.2	78.2	192	205
MOT16-05	43.2	43.5	73.8	73.9	78	85
MOT16-09	54.2	54.4	79.7	79.7	35	32
MOT16-10	36.5	37.0	73.7	73.8	133	111
MOT16-11	45.8	46.2	81.2	81.2	25	25
MOT16-13	28.6	28.67	72.7	72.7	77	79
Combined	40.0	41.0	76.9	76.9	647	648

Note: With BG = Original Deep SORT using the original bounding boxes. Without BG = Proposed Deep SORT with segmentation-guided background removal. $\uparrow$ indicates higher values are better, whereas $\downarrow$ indicates lower values are better.

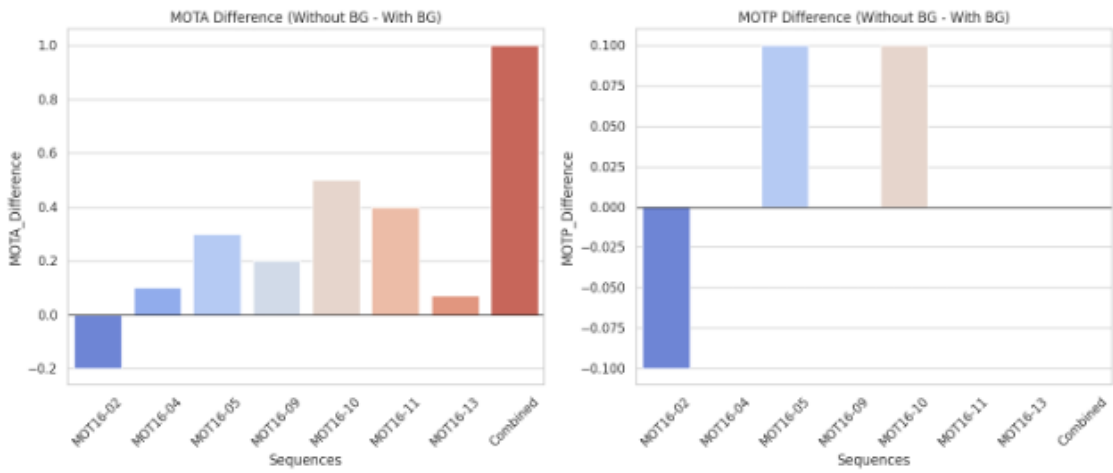


Figure 3. MOTA and MOTP difference results of various sequences.

Table 2. Additional Tracking Performance Comparison on MOT16 Sequences

Sequence	Fragmentations ↓		Mostly Tracked ↑		Total Detections ↑	
	With BG	Without BG	With BG	Without BG	With BG	Without BG
MOT16-02	239	240	8	8	9109	9146
MOT16-04	554	555	13	13	30560	30562
MOT16-05	167	164	29	33	5851	5918
MOT16-09	78	73	11	11	4386	4404
MOT16-10	328	322	11	11	9265	9156
MOT16-11	78	78	21	22	7860	7848
MOT16-13	268	270	22	21	6656	6692
Combined	1712	1702	115	119	73687	73726

Note: With BG = Original Deep SORT using the original bounding boxes. Without BG = Proposed Deep SORT with segmentation-guided background removal. ↑ indicates higher values are better, whereas ↓ indicates lower values are better.

As shown in Table 1, this architecture achieves 1% improvement in overall tracking accuracy, with an increase in MOTA. The MOTP metric remains unchanged, indicating that localization precision is not affected by this architecture. Identity switches exhibits inconsistency; while some sequences show marginal degradation, others show improvement. This illustrates that although the architecture improves feature representation in certain cases, it does not fully solve the problem of identity switches in crowded scenes.

Fig. 3 exhibits the sequence-wise differences in MOTA and MOTP. This architecture improves MOTA for MOT16 sequences with gains in all but MOT16-02, which exhibits a slight decrease in performance. This variation is expected because in challenging sequences, imperfect segmentation masks fail to accurately separate foreground from background and may remove some useful features. Improved association and higher MOTA are observed for sequences containing greater background clutter because they benefit more from background removal. While minor variations are observed in MOTP, it remains unchanged across most sequences. Overall, our proposed method shows significant improvement. Imperfect segmentation can be reduced if Mask R-CNN is finetuned on the tracking MOT16 dataset, which can potentially result in consistent improvement across all metrics.

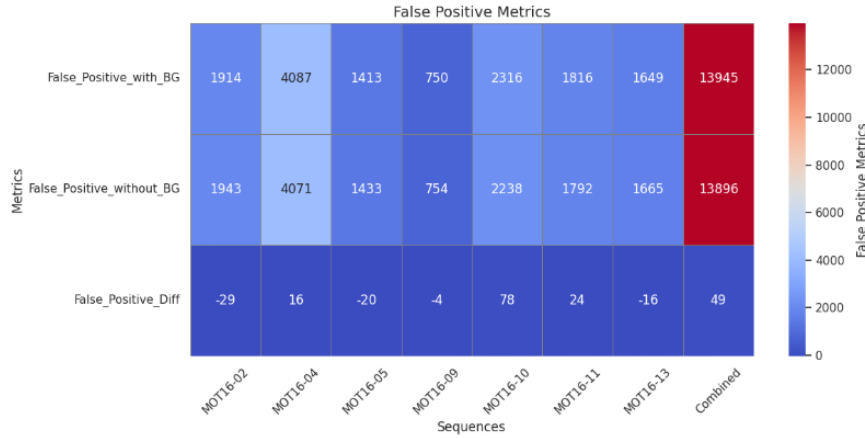
**Figure 4.** False positives for different sequences.

Fig. 4 and 5 show reductions in false positives (FP) and false negatives (FN), indicating improved tracking performance. An FP occurs when a tracking system incorrectly detects an object that is not present. Similarly, an FN occurs when the tracking system fails to detect an object that is present in the scene. These improvements are not consistent in all the sequences because some show a slight increase in these metrics. However, the overall reduction in these metrics in combined results suggests that segmentation-based background removal improves the robustness of data association within the Deep SORT framework.

Table 2 indicates that our proposed method results in a lower number of fragments, signifying more consistent data association between consecutive detections. The enhanced appearance descriptor for identity

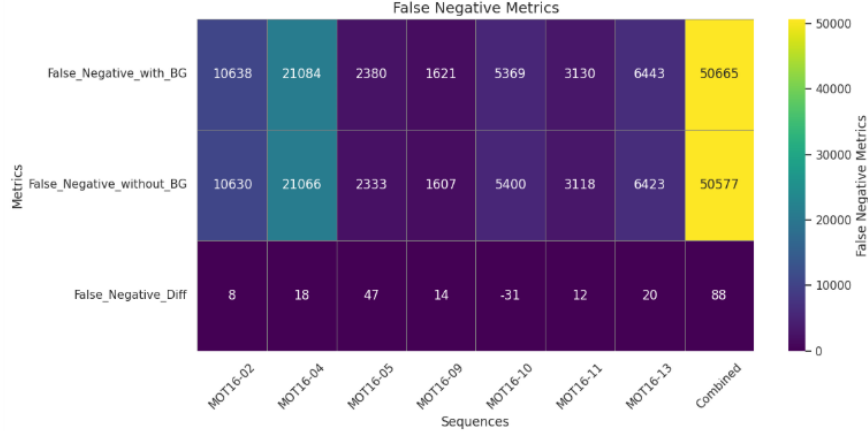


Figure 5. False negatives for different sequences.

matching due to background subtraction results in fewer fragmented tracks. The increased number of mostly tracked objects and increase in total detections indicate that object trajectories are preserved more successfully. These metrics are valuable for long video sequences, particularly in traffic monitoring and surveillance applications. These results are promising, as they indicate that tracking can be improved without modifying the underlying Deep SORT framework. It is important to note that while these metrics suggest improvements, due to the domain difference between the MOT16 challenge and the COCO2017 dataset, the generated segmentation masks are suboptimal. Thus, the performance remains constrained in high occluded and dense scenes.

The proposed framework adds extra computational complexity in the inference stage of Mask R-CNN before appearance feature extraction. The tracking pipeline (Deep SORT) is not changed, and the extra computation is only for foreground segmentation. Thus, the approach proposed is a compromise between the appearance feature quality and computational cost. Computational efficiency was not the primary focus in this work, but will be explored in future research to reduce the extra inference overhead without compromising tracking performance while using lightweight segmentation models.

4. Limitations

Mask R-CNN is pre-trained on the COCO2017 dataset, which contains a variety of object classes, including the human class. But due to the domain differences, such as crowd concentration, environmental factors, and object density between COCO2017 and MOT16 challenge, the performance of the proposed framework is limited on the MOT16 challenge. This is because without ground-truth annotations of MOT16 dataset, the Mask RCNN model cannot be finetuned to generate accurate masks for humans or other objects in the MOT16 dataset, resulting in suboptimal segmentation masks, often resulting in incomplete subtraction of background clutter. Consequently, this loss of appearance information, in some sequences, negatively affects feature quality, increasing the probability of identity switches and reducing overall accuracy in detection and tracking tasks.

The proposed method is compared only against the original Deep SORT model. A comparison with recent trackers such as ByteTrack and BoT-SORT to evaluate the generalizability of the proposed research is an important direction for future research. Furthermore, the reported results are obtained with a single experimental run, which was used according to the standard Deep SORT evaluation protocol. There was no repetition of experiments with several runs, different random seeds, or statistical significance analysis. Future work will be conducted with repeated evaluations and statistical analysis to further evaluate the robustness and reproducibility of the proposed method.

5. Conclusion

Algorithms for MOT often suffer degradation and increased FP and FN due to constraints such as occlusion, crowded scenes, and similar object appearance. We have proposed a framework to mitigate the limitations by integrating Mask R-CNN for object detection and segmentation with Deep SORT for multiple object tracking. Mask R-CNN refined detections by suppressing background information within bounding boxes and improving appearance quality for data association. The proposed framework exhibited reduced false positives, false negatives, and fragmented tracks with an improvement in MOTA when evaluated on MOT16 challenge. We also highlight that segmentation-based tracker, trained on the COCO dataset exhibits limitations when applied to the MOT16 dataset. This issue arises because the COCO dataset is relatively sparse in terms of human annotations, leading to suboptimal mask generation for persons in the MOT16 dataset. Hence, training Mask R-CNN directly on the MOT dataset can be a potential direction for future work. For training Mask R-CNN effectively, the MOT dataset needs pixel-level segmentation masks. Generating such annotations would enable effective retraining, enhanced detection performance, and improve mask quality. Furthermore, Deep SORT is chosen as a standard algorithm to prove the concept of our idea. This technique can easily be extended to recent detection-based multiple object tracking algorithms.

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